Droplet sedimentation, precipitation suppression and resolution in the Nd–LWP response of DP-EAMxx (DYCOMS-II RF02)

Technical report, version 3

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Abstract

We ran DP-EAMxx on the DYCOMS-II RF02 nocturnal drizzling stratocumulus for 12 h. The matrix covers grid spacing Δx from 40 km to 312 m, droplet number Nd from 20 to 1000 cm^{-3}, a cloud-droplet sedimentation multiplier (×0.5, 1, 2), precipitation suppression (KK2000 Nd dependence on or off) and, below 1 km, 3D turbulence on or off. In all, 335 runs are complete and analysed. That includes all 36 runs at 312 m with 3D turbulence off, and 11 of the 36 with it on (68 % of 312 m model-hours). The Nd–LWP response depends strongly on resolution, and the two microphysical pathways trade places as Δx shrinks. At ≥10 km, LWP rises monotonically with Nd, and precipitation suppression explains all of that rise. As Δx shrinks, entrainment increases and depends more strongly on Nd, the precipitation-suppression gain collapses (from +134 $\mathrm{g\,m}^{-2}$ at 40 km to +6 $\mathrm{g\,m}^{-2}$ at 312 m, both at Nd 1000), and the LWP maximum moves to lower Nd (Nd > 1000 at ≥10 km, 500 at 5 km, 200 at 2.5 km–625 m, 100 at 312 m). The result is an inverted V whose falling branch is entrainment-driven: it persists with precipitation suppression off, and faster sedimentation strengthens it. The low-Nd branch ($S_{20\to50}$) barely changes with resolution below 2.5 km, while LWP itself has not converged at 312 m. 3D turbulence changes LWP by less than 1.5 $\mathrm{g\,m}^{-2}$ at both 625 m and 312 m.

Key results

1. **The inverted V forms and its peak moves to lower Nd as Δx shrinks** (Figs. 1, 2; Table 1). With sed×1 and suppression on, LWP peaks at Nd ≥1000 at ≥10 km, at 500 at 5 km, at 200 from 2.5 km to 625 m, and at 100 at 312 m. The high-Nd susceptibility $S_{100\to200}$ falls monotonically: 0.27 (40 km), 0.20 (5 km), 0.11 (2.5 km), 0.07 (1.25 km), 0.03 (625 m), −0.05 (312 m) (Fig. 3).
2. **Precipitation suppression fades with resolution** (Fig. 4a). At high Nd, LWP(on) − LWP(off) is +134 (40 km), +58 (5 km), +34 (2.5 km), +27 (1.25 km), +16 (625 m) and +6 $\mathrm{g\,m}^{-2}$ (312 m), all at Nd 1000. With suppression off, S is near zero or negative at every resolution.
3. **The falling branch comes from entrainment and is strengthened by sedimentation.** It survives with suppression off (at 312 m, $S_{100\to200}$ is −0.07 off and −0.05 on), while w_e keeps rising with Nd (Fig. 1). With suppression on it steepens with the sed factor (625 m: $S_{200\to1000} = -0.02/-0.04/-0.06$ for sed ×0.5/1/2; 312 m: $-0.02/-0.04/-0.07$ on and $-0.02/-0.05/-0.06$ off).
4. **Sedimentation–entrainment** (Fig. 5). At ≥10 km, w_e switches between two levels (about 4.0 and 3.0 $\mathrm{mm\,s}^{-1}$) at a cloud-top fall-speed proxy near 3.5. At ≤5 km, w_e falls smoothly with the proxy, and at a given proxy it is higher the finer the grid.
5. **The low-Nd branch is robust; LWP is not converged.** Below 5 km, $S_{20\to50}$ (suppression on) is 0.56–0.60, 0.65–0.76 and 0.88–1.00 for sed ×0.5/1/2, with little dependence on Δx. LWP, cloud cover and w_e still change strongly between 625 m and 312 m (Fig. 6): at Nd 100, LWP goes from 120 to 76 $\mathrm{g\,m}^{-2}$, cloud cover from 0.92 to 0.83 and w_e from 7.0 to 8.1 $\mathrm{mm\,s}^{-1}$, and the cells shrink (Fig. 7).
6. **3D turbulence makes almost no difference at 625 m or 312 m** (Fig. 9). At 625 m (36 pairs), $|\Delta\mathrm{LWP}| < 1.5\ \mathrm{g\,m}^{-2}$ and $|\Delta\text{cloud cover}| < 0.007$. At 312 m (11 pairs so far, Nd 20–100), $|\Delta\mathrm{LWP}| \le 1.0\ \mathrm{g\,m}^{-2}$ and $|\Delta\text{cloud cover}| \le 0.006$. The one consistent signal is w_e, which is 0.02–0.07 $\mathrm{mm\,s}^{-1}$ lower (<1 %).

7. **Steadiness** (Fig. 10). At ≤2.5 km, $|d\mathrm{LWP}/dt| \leq 4.5\,\mathrm{g\,m^{-2}\,h^{-1}}$ over hours 10–12. At 40–10 km, LWP is still rising at high Nd with suppression on, by up to $+18\,\mathrm{g\,m^{-2}\,h^{-1}}$.

Table 1: LWP ($\mathrm{g\,m^{-2}}$, hours 10–12) with sed×1 and 3D turbulence off. Bold marks the maximum with suppression on.

Δx	suppression	Nd 20	50	100	200	500	1000
40 km	on	135	188	246	295	333	**342**
	off	234	221	216	211	209	208
10 km	on	134	186	240	281	308	**312**
	off	232	217	213	208	206	204
5 km	on	82	137	184	214	**222**	218
	off	135	149	153	158	159	160
2.5 km	on	67	119	163	**177**	175	170
	off	117	137	139	138	137	136
1.25 km	on	65	111	149	**157**	154	149
	off	116	127	127	124	122	122
625 m	on	52	98	120	**122**	119	115
	off	103	109	108	103	101	99
312 m	on	40	72	**76**	74	72	70
	off	79	79	72	69	66	64

1 Setup

Case and model. DYCOMS-II RF02 nocturnal drizzling stratocumulus (shortwave off; T, q nudged above 650 hPa; IOP surface fluxes and subsidence). The forcing is constant over 0–12 h. We run doubly periodic EAMxx (SHOC, P3, SPA, RRTMGP; 128 levels; stub sea ice). Domains are 600 km at 40 km, 300 km at 20 km, and 240 km from 10 km down. Time steps follow the earlier ACI study: the physics step is 100 s at ≥5 km and 75/60/30/15 s at 2.5/1.25/0.625/0.3125 km. The E3SM branch is `mahf708/eamxx/p3-cloud-sed-factor` (E3SM PR #9017 plus a P3 sedimentation knob).

Matrix. The full matrix is Δx (40, 20, 10, 5, 2.5, 1.25 km, 625, 312 m) × Nd target (20, 50, 100, 200, 500, 1000 $\mathrm{cm^{-3}}$) × `cloud_sedimentation_factor` (0.5, 1, 2) × precipitation suppression (on, off), plus 3D turbulence (off, on) at 625 and 312 m.

- `p3::cloud_sedimentation_factor` is a new P3 option that scales the mass- and number-weighted cloud-droplet fall speeds, like MG2's `cld_sed`.
- Precipitation suppression off means KK2000 autoconversion with the Nd exponent set to 0 and the prefactor set to $1350 \cdot 63^{-1.79} = 0.812$. That matches the default rate at Nd 63.
- The SPA activation factor sets Nd. At ≥625 m the diagnosed in-cloud Nd is 10–25 % above target; at 312 m it is 15–20 % below.
- The 625 and 312 m runs are done in restart segments across interactive, debug and regular jobs. We checked that segmented runs are bit-for-bit identical to straight runs.

Diagnostics.

- Values are means over hours 10–12 unless stated otherwise. Cloud cover is the fraction of columns with LWP $> 10\,\mathrm{g\,m^{-2}}$.
- $w_e = dz_i/dt - w_{\mathrm{subs}}(z_i)$.
- $S = d\ln \mathrm{LWP}/d\ln N_d$, computed from the diagnosed Nd.
- The LWP spatial variance is computed online with the expression syntax, and cloud-top fields with the new `aerocom(..)` operator.
- The 6 h results (technical note v1) are consistent with the first 6 h of these runs, which reproduce them bit-for-bit.

2 Results

2.1 Nd response at every resolution

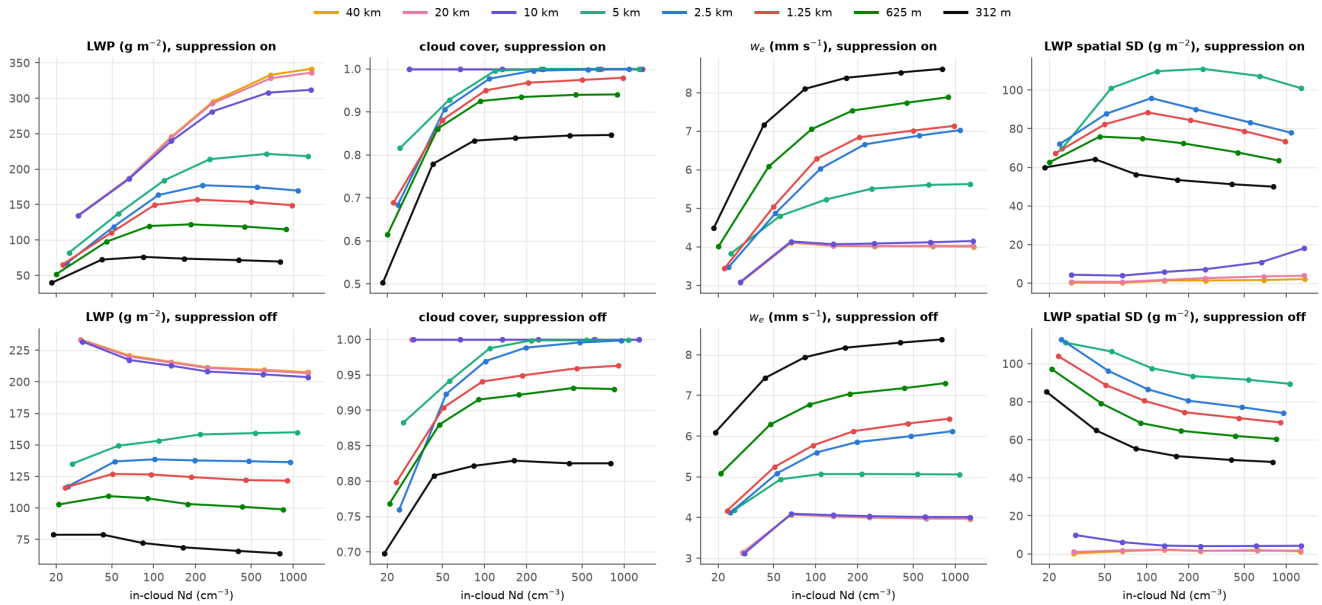


Figure 1: Nd response at sed×1, all resolutions on shared axes. Top row: suppression on. Bottom row: suppression off.

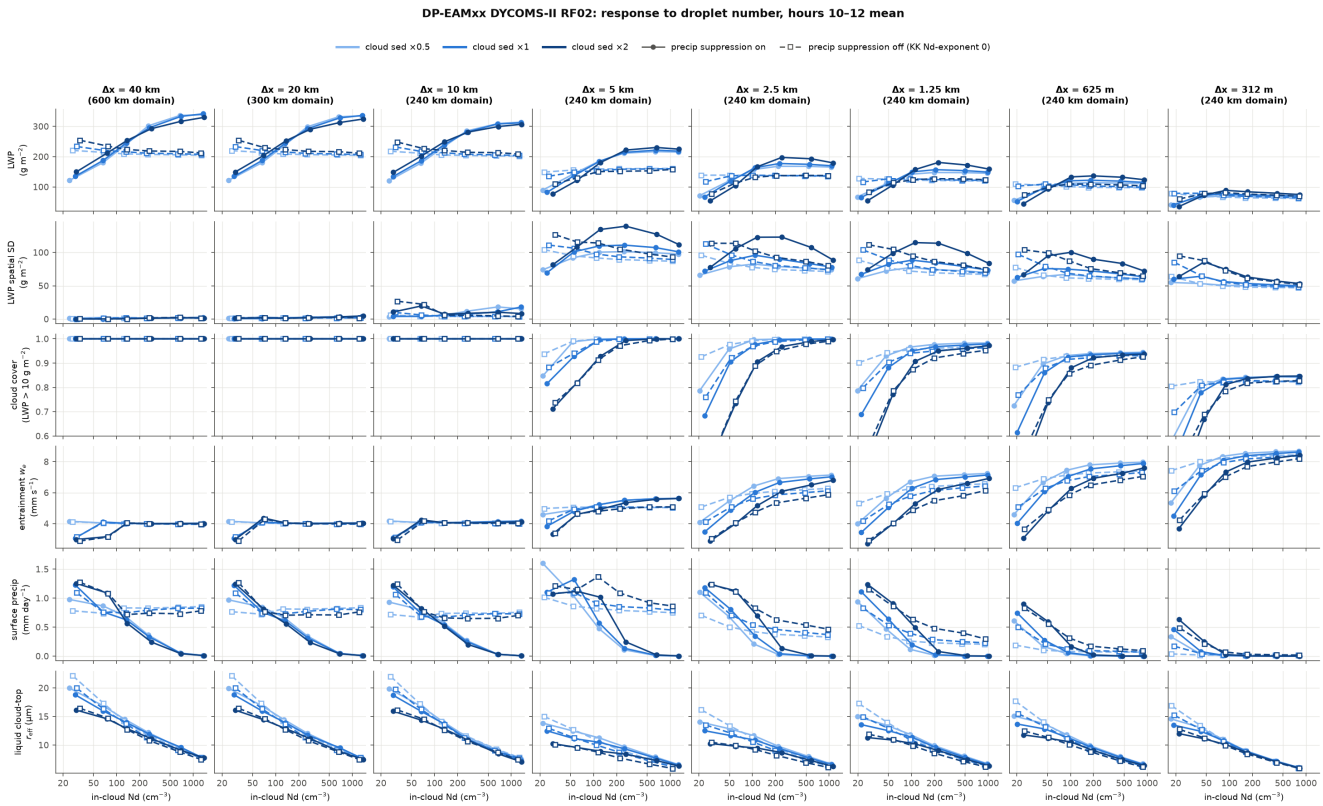


Figure 2: Every resolution, sed factor and suppression setting. Shade: sed factor. Solid lines: suppression on. Dashed lines: off.

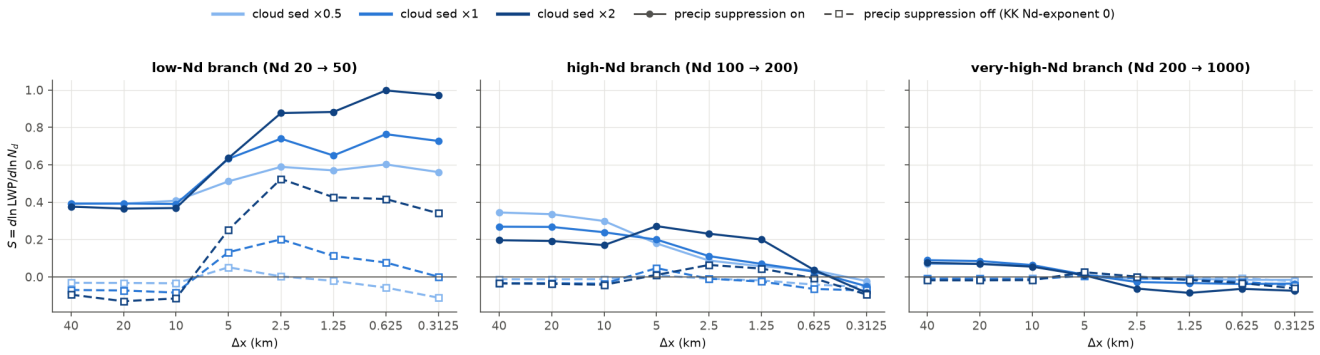


Figure 3: Susceptibility on three Nd branches against Δx.

2.2 Pathways

The pathway figure (Fig. 4) takes the difference between runs that differ in one switch only. The precipitation-suppression term crosses zero near Nd 63 by construction. Above Nd 100 it falls off steeply with Δx. The sedimentation term changes sign between the coarse and fine regimes.

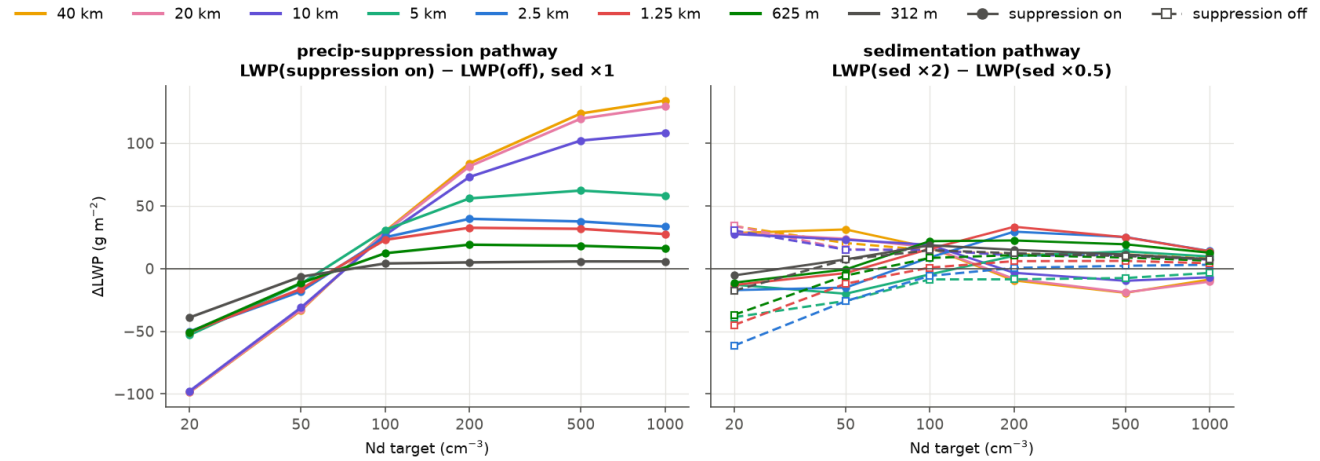


Figure 4: (a) Precipitation-suppression pathway (sed×1). (b) Sedimentation pathway, with suppression on (solid) and off (dashed). Colour: resolution.

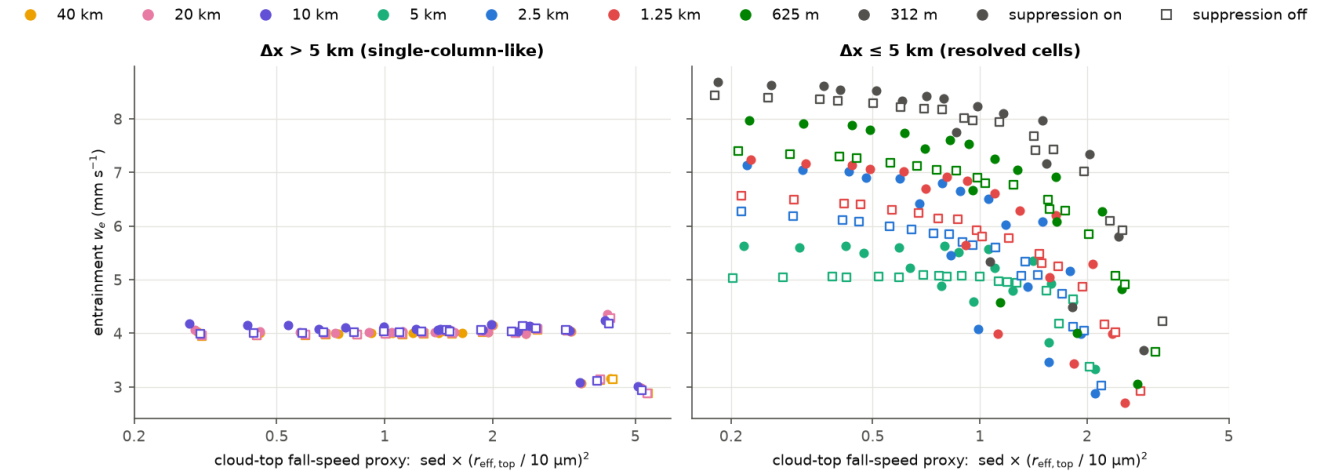


Figure 5: w_e against the cloud-top fall-speed proxy sed$\times(r_{\mathrm{eff,top}}/10\,\mu\mathrm{m})^2$, for every run with 3D turbulence off.

2.3 Resolution dependence and cloud morphology

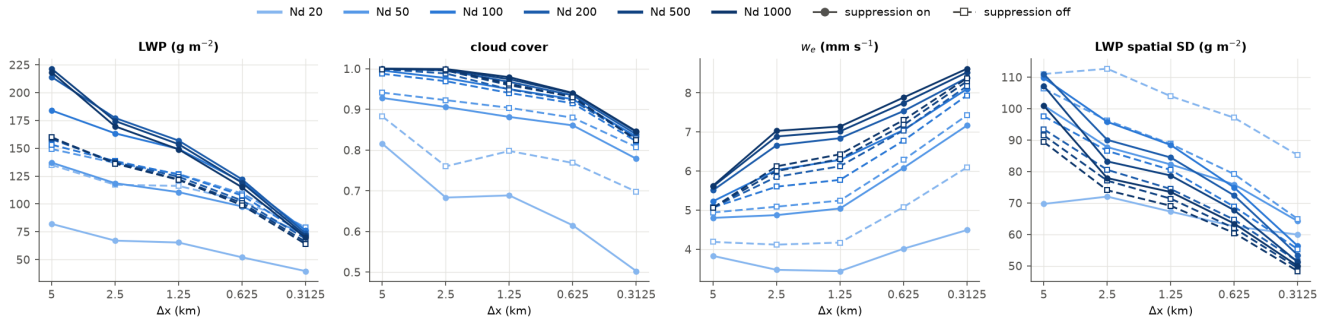


Figure 6: Hour 10–12 means against Δx ($\leq$5 km), sed×1, for every Nd.

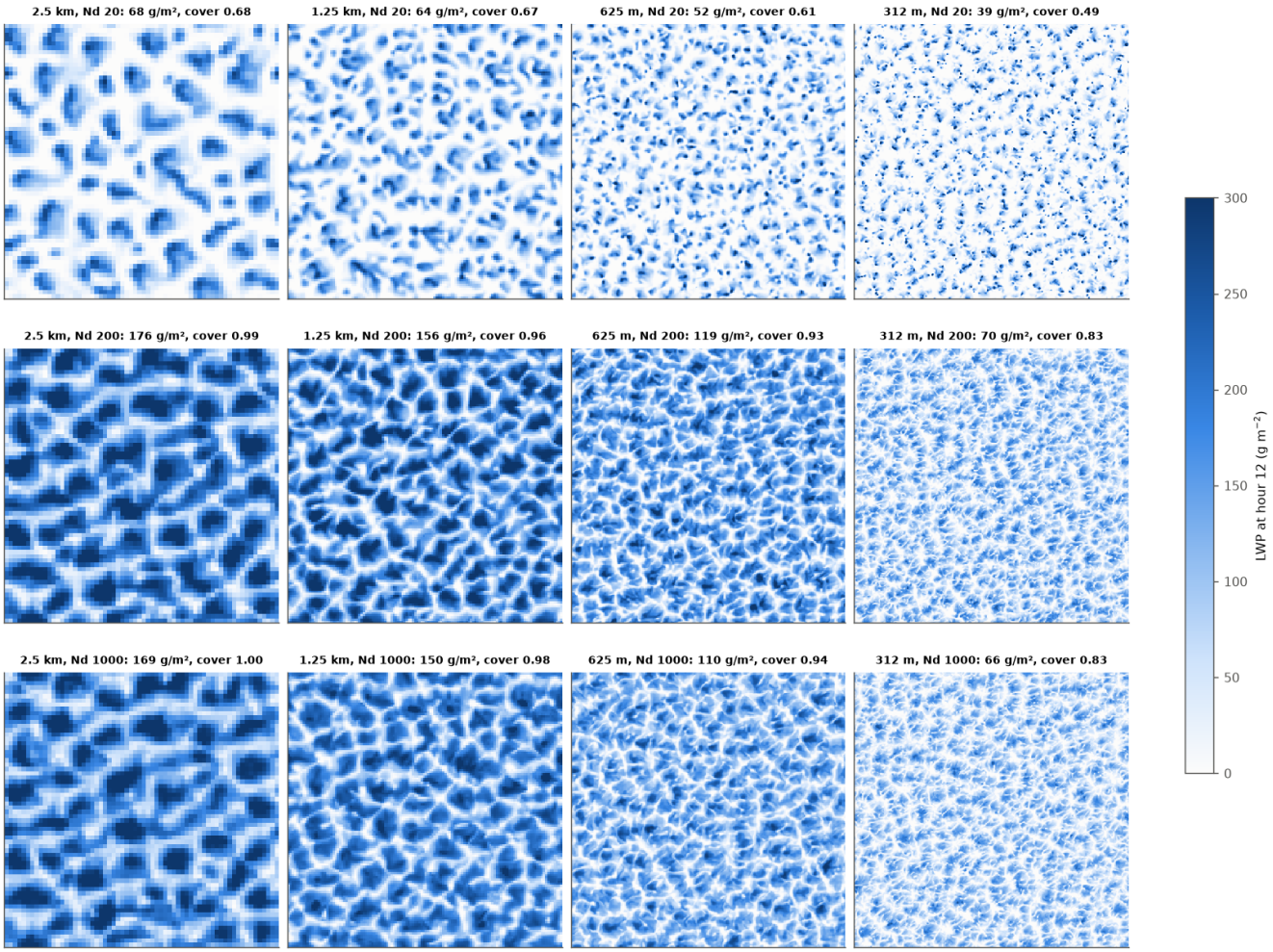


Figure 7: LWP at hour 12 over the full 240 km domain, sed×1, suppression on. Rows: Nd 20/200/1000. Columns: 2.5 km–312 m.

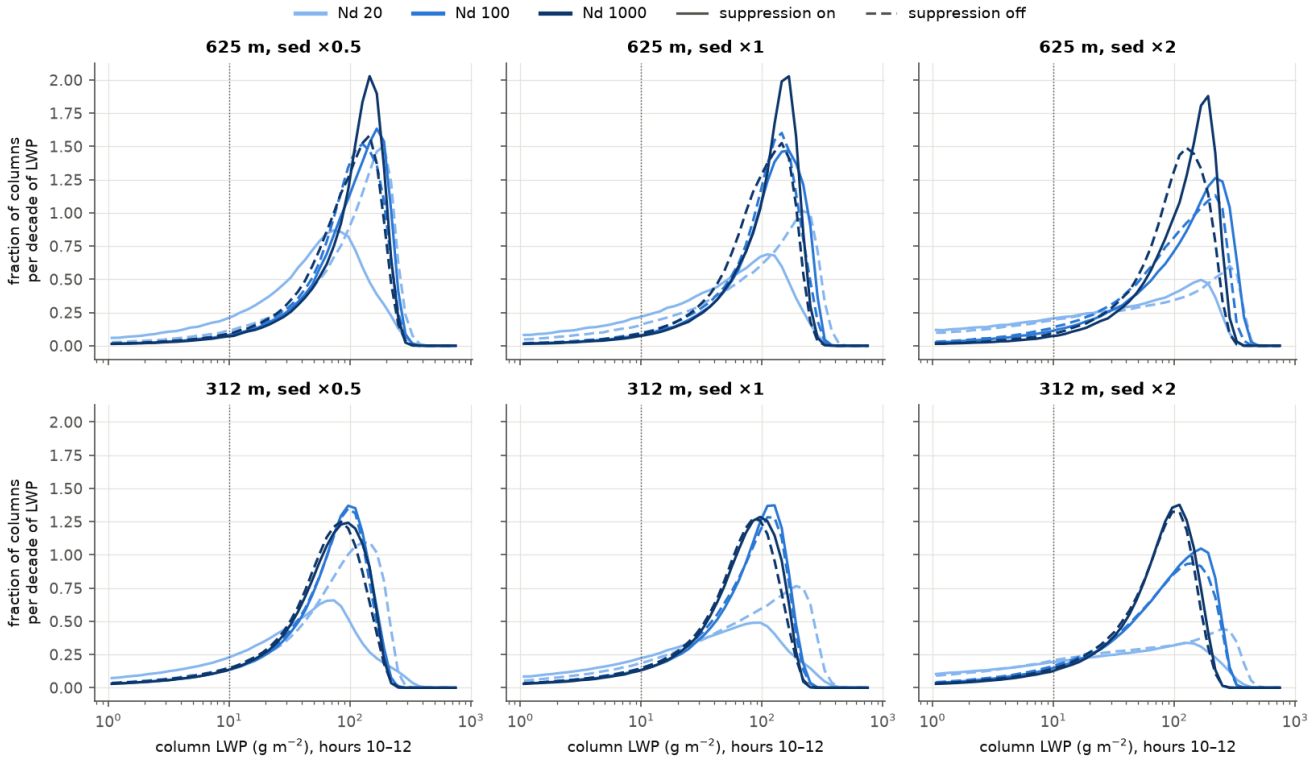


Figure 8: Column-LWP distributions over hours 10–12 at 625 m and 312 m.

2.4 3D turbulence

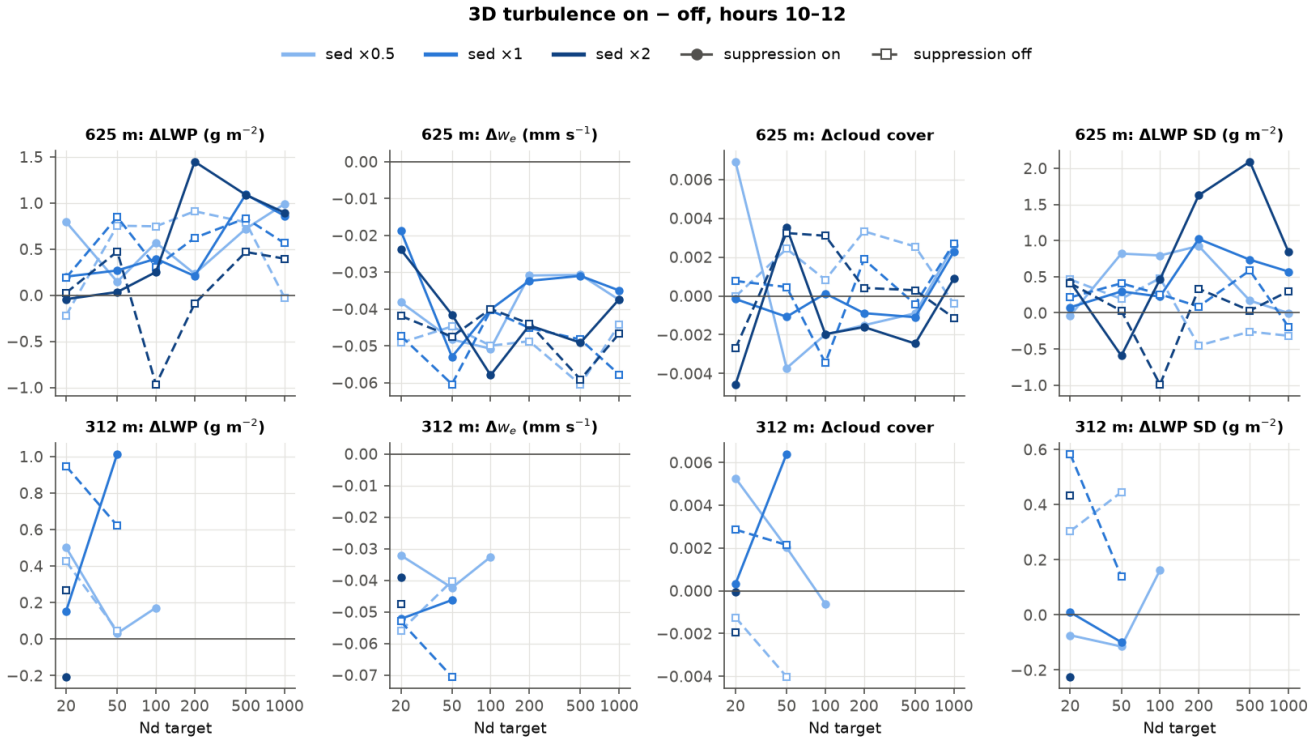


Figure 9: 3D turbulence on minus off at 625 m (all 36 pairs) and at 312 m (the 11 pairs complete so far).

2.5 Transience

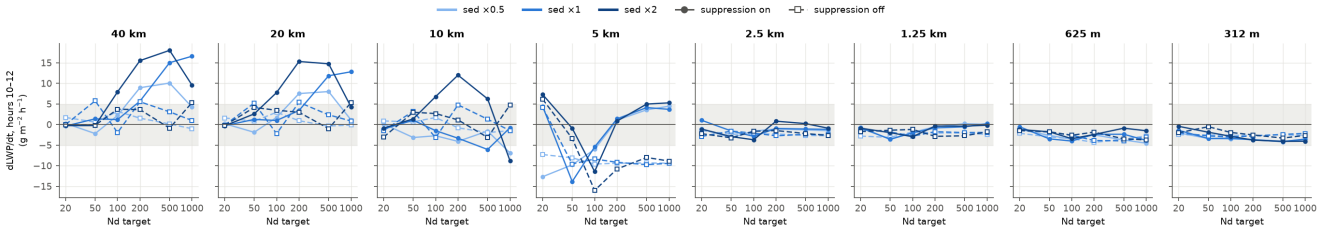


Figure 10: Linear LWP trend over hours 10–12. The grey band marks $\pm 5\,\mathrm{g\,m^{-2}\,h^{-1}}$.

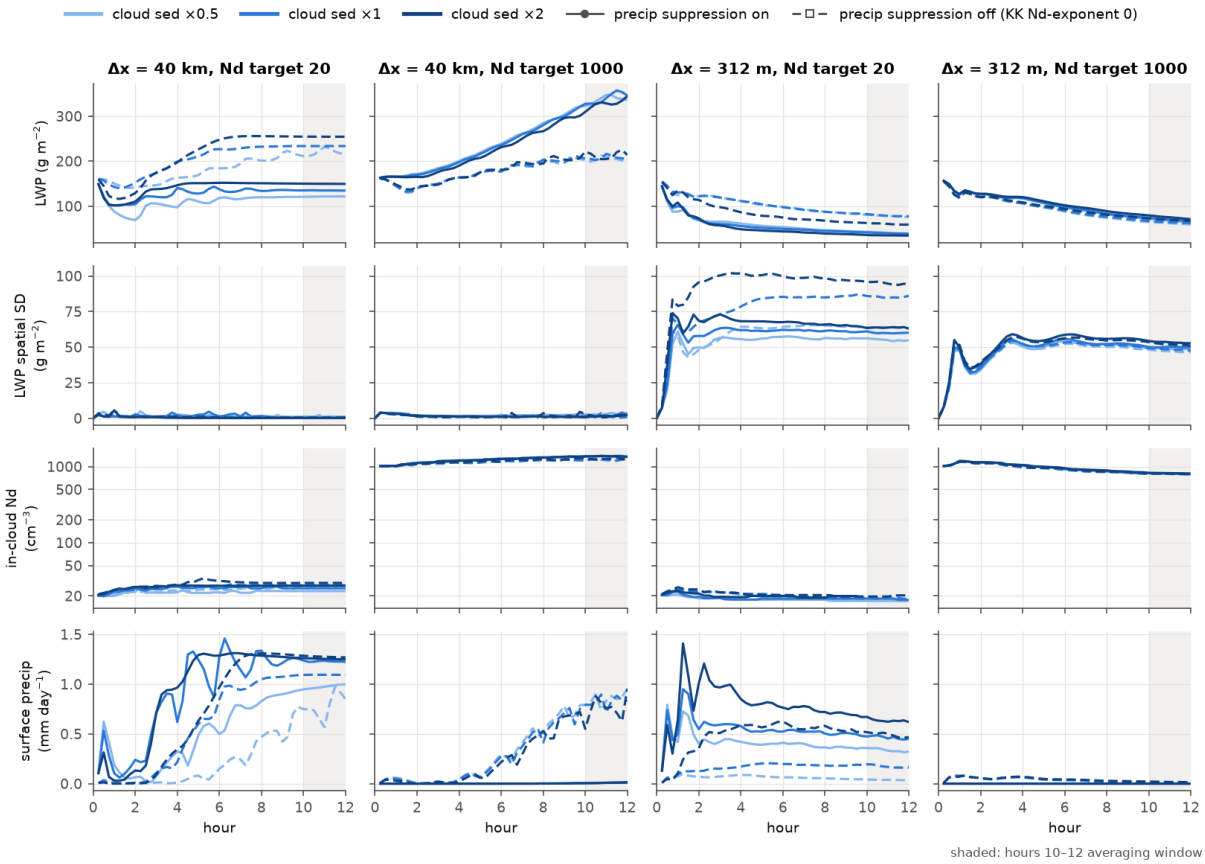


Figure 11: Time series at 40 km and 312 m for Nd 20 and 1000.

3 Caveats, status and next steps

- **312 m is 68 % done** (589 of 864 model-hours).
  - Every run with 3D turbulence off is complete. Of the 36 runs with it on, 25 are still to come.
  - Two 16-node regular jobs are expected to start this morning; the rest should be done within about a day.
  - A final version of this report will follow.
- **No convergence.** The step from 625 m to 312 m is as large as any coarser step. 156 m (or a smaller domain at 156 m) is the open question for discussion.
- **Single realization per run.** Seed ensembles at the km-scale and sub-km high-Nd runs would put error bars on the small negative S values.

- **Diagnostics.**
  - Surface precipitation includes cloud-droplet sedimentation.
  - w_e comes from a linear fit of z_i.
  - The fall-speed proxy stands in for the actual cloud-top sedimentation flux.

4 Data and code (Perlmutter, world-readable)

- 12 h runs: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/sedps_<res>_nd<Nd>_sed<0p5|1|2>_ps<on|off>[_t3don]/run/`
- This report's analysis: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/analysis_report3/`
  - `summary_h10-12.csv` and `timeseries_15min.csv`: one row per run (and per 15-min step), with columns for the run settings including `turb3d`.
  - `susceptibility.csv`, `lwp_trend_h10-12.csv`, `turb3d_on_minus_off.csv` and all figures.
- 6 h runs and technical note v1: `/pscratch/sd/m/mahf708/dp_screamxx_sedps/`
- Scripts, R tools and LaTeX source: `/pscratch/sd/m/mahf708/scmlib/sed-precsupp-study/`
  - In R: `source(".../R/sedps.R"); sedps_runs("/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h", hours = 12)`.
- Code: E3SM `mahf708/eamxx/p3-cloud-sed-factor` and scmlib `sed-precsupp-study`.